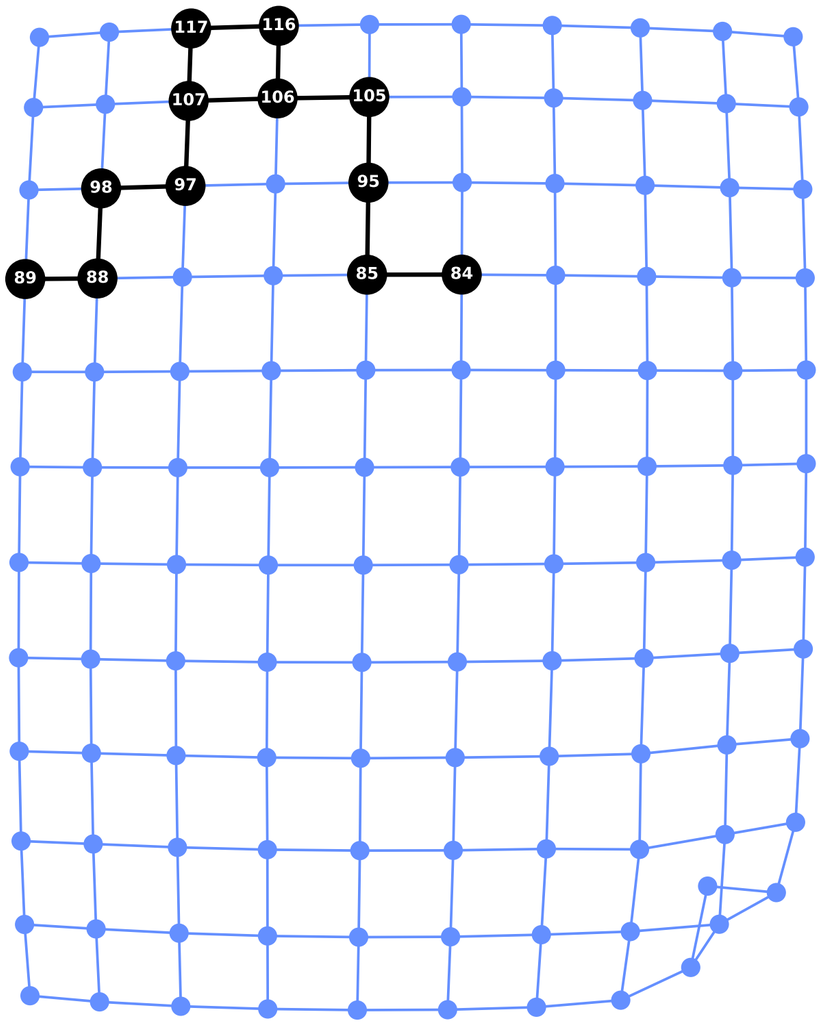
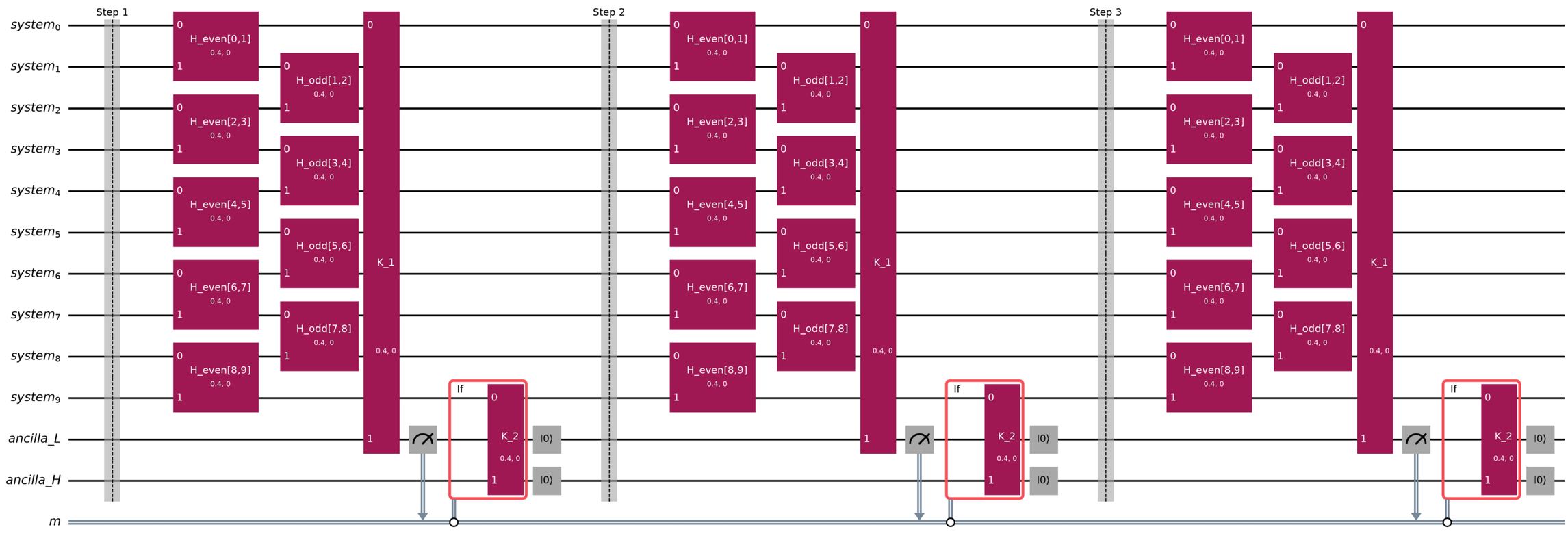
Dynamic Lie-Trotter layout on ibm_phoenix: 10 system qubits + 2 ancillas



Initial logical-to-physical mapping

Logical	Role	Physical
q[0]	system[9]	85
q[1]	system[8]	95
q[2]	system[7]	105
q[3]	system[6]	106
q[4]	system[5]	116
q[5]	system[4]	117
q[6]	system[3]	107
q[7]	system[2]	97
q[8]	system[1]	98
q[9]	system[0]	88
q[10]	ancilla_L (low)	89
q[11]	ancilla_H (high)	84

3 dynamic Lie-Trotter steps before transpilation (dt=0.2)



system[i] = model site i = original logical q[9-i]. ancilla_L/H are the low/high shared-ancilla encoding qubits.
Black device nodes show physical indices. Table gives initial placement; routing SWAPs can change placement later.
Lower panel is the matching compact circuit, not its decomposed hardware gate sequence.